

FRIDAY

September 18, 2026

DO NOW

1. Put your maze in the bin
2. Take out your 2.4 Practice

AGENDA

- Mini-quiz
- 2.5 Notes (solving)
- Practice

WHAT YOU NEED

- Pencil
- Notebook

Solve each equation. Check your solution.

1. $5x - 3 = 13 - 3x$ **2**

2. $-4r - 11 = 4r + 21$ **-4**

3. $1 - m = 6 - 6m$ **1**

4. $14 + 5n = -4n + 17$ **$\frac{1}{3}$**

5. $\frac{1}{2}k - 3 = 2 - \frac{3}{4}k$ **4**

6. $\frac{1}{2}(6 - y) = y$ **2**

7. $3(-2 - 3x) = -9x - 4$ **no solution**

8. $4(4 - w) = 3(2w + 2)$ **1**

9. $9(4b - 1) = 2(9b + 3)$ **$\frac{5}{6}$**

10. $3(6 + 5y) = 2(-5 + 4y)$ **-4**

11. $-5x - 10 = 2 - (x + 4)$ **-2**

12. $6 + 2(3j - 2) = 4(1 + j)$ **1**



13. $\frac{5}{2}t - t = 3 + \frac{3}{2}t$ no solution

14. $1.4f + 1.1 = 8.3 - f$ 3

15. $\frac{2}{3}x - \frac{1}{6} = \frac{1}{2}x + \frac{5}{6}$ 6

16. $2 - \frac{3}{4}k = \frac{1}{8}k + 9$ -8

17. $\frac{1}{2}(3g - 2) = \frac{g}{2}$ 1

18. $\frac{1}{3}(n + 1) = \frac{1}{6}(3n - 5)$ 7

19. $\frac{1}{2}(5 - 2h) = \frac{h}{2}$ $1\frac{2}{3}$

20. $\frac{1}{9}(2m - 16) = \frac{1}{3}(2m + 4)$ -7

21. $3(d - 8) - 5 = 9(d + 2) + 1$ -8

22. $2(a - 8) + 7 = 5(a + 2) - 3a - 19$ all numbers





23. NUMBERS Two thirds of a number reduced by 11 is equal to 4 more than the number. Find the number. **-45**

24. NUMBERS Five times the sum of a number and 3 is the same as 3 times the difference of twice the number and 1. What is the number? **18**

25. NUMBER THEORY Tripling the greater of two consecutive even integers gives the same result as subtracting 10 from the lesser even integer. What are the integers? **-8, -6**

26. GEOMETRY The formula for the perimeter of a rectangle is $P = 2\ell + 2w$, where ℓ is the length and w is the width. A rectangle has a perimeter of 24 inches. Find its dimensions if its length is 3 inches greater than its width.
4.5 in. by 7.5 in.



2.3-2.4 Mini-Quiz

When you are finished:

1. Hand in your quiz
2. Take today's notes
3. Go on google classroom and complete the notes
4. Show me your completed notes
5. 2.5 Skills Practice WS



10:00



2.5 Absolute Value Equations

**SWBAT solve absolute value equations
by solving 2 cases**

What is absolute value?

The **absolute value** of a number represents its distance from zero on the number line.

$$|4| = 4 \qquad |-4| = 4$$

What are the possible values for $|x| = 5$

5 and -5

Steps for solving an absolute value equation

1. **Isolate** the absolute value on the left side
2. Drop the absolute value bars and create **two cases**, one positive and one negative.
3. **Solve** both cases.
4. Write your **solution set**.
5. **Graph** your solution if required.

Steps:

$$|f + 2| = 4$$

1. Isolate the absolute value on the left side.

2. Drop the absolute value bars and create two cases, one positive and one negative.

3. Solve both cases.

4. Write your solution set.

5. Graph your solution if required.

Case 1

$$f + 2 = 4$$

$$\begin{matrix} -2 & -2 \\ f & = 2 \end{matrix}$$

$$f = 2$$

Case 2

$$f + 2 = -4$$

$$\begin{matrix} -2 & -2 \\ f & = -6 \end{matrix}$$

$$f = -6$$

Solution set is written in
braces listed least to
greatest $\longrightarrow \{-6, 2\}$



Steps:

1. Isolate the absolute value on the left side.
2. Drop the absolute value bars and create two cases, one positive and one negative.
3. Solve both cases.
4. Graph your solution if required.

$$|x + 4| + 3 = 17$$

-3 -3

$$|x + 4| = 14$$

Case 1

$$\begin{aligned}x + 4 &= 14 \\-4 &\quad -4 \\x &= 10\end{aligned}$$

Case 2

$$\begin{aligned}x + 4 &= -14 \\-4 &\quad -4 \\x &= -18\end{aligned}$$

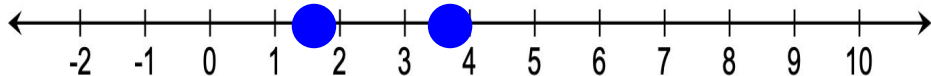
$\{-18, 10\}$



Steps:

1. Isolate the absolute value on the left side.
2. Drop the absolute value bars and create two cases, one positive and one negative.
3. Solve both cases.
4. Graph your solution if required.

$\{5/4, 15/4\}$



$$3 | 4w - 10 | - 5 = 10$$

$+5 \qquad +5$

$$\frac{3 | 4w - 10 |}{3} = \frac{15}{3}$$

$$| 4w - 10 | = 5$$

Case 1

$$4w - 10 = 5$$

$$+10 \quad +10$$

$$4w = 15$$

$$\frac{4}{4}$$

$$\frac{15}{4}$$

$$x = 15/4$$

Case 2

$$4w - 10 = -5$$

$$+10 \quad +10$$

$$4w = 5$$

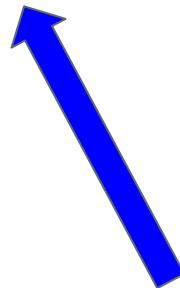
$$\frac{4}{4}$$

$$\frac{5}{4}$$

$$x = 5/4$$

$$|x - 1| = -3$$

What do you notice about this equation?



Distance is never negative for absolute value.

This equation means the distance between x and 1 is -3.
Since distance cannot be negative, the solution is the empty set $\emptyset$.

No Solution!

Video Help

Skills Practice